

Every Sobolev Gaussian measure has a measurable Gaussian-process realization

Automated Math Discovery Run `fa_banach_001`, lane 8

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Result. Concluding question (ii) of Henderson has an affirmative answer. Every Gaussian measure on $W^{m,p}(\mathcal{D})$, for integer $m \geq 0$ and $1 \leq p < \infty$, is induced by a jointly measurable Gaussian process whose sample classes are $W^{m,p}$ -valued almost surely. The proof also applies to any separable Banach function space continuously injected into an L^p space.

1 Source questions

The paper [1] asks:

1. whether Sobolev small-ball problems can be handled using spectral properties of the covariance, such as its nuclear norm;
2. whether every Gaussian measure on $W^{m,p}(\mathcal{D})$ is induced by some Gaussian process.

It notes that its Proposition 2.9 gives the second statement for $m = 0$, that is, on $L^p(\mathcal{D})$. We completely answer (ii), and give exact limitations on the nuclear-norm suggestion in (i).

here to PDE-related machine learning problems is certainly an interesting continuation of the results of this article.

The following directions are interesting for generalizing the results presented here. First, similar spectral/integral criteria should be obtained for fractional Sobolev and Besov spaces. Second, similar results should be sought to tackle the cases $p = 1$ and $p = +\infty$. Linked to the case $p = 1$, results should be sought for the space of functions of bounded variations ([7, p. 269]), which are important in many problems related to physics. The following open questions are also relevant for Gaussian process theory: (i) Can the small ball problem for Gaussian processes whose sample paths lie in a Sobolev spaces be tackled only using spectral properties of the covariance operator, e.g. its nuclear norm? (ii) Are all Gaussian measures over $W^{m,p}(\mathcal{D})$ induced by some Gaussian process? Proposition 2.9 shows that it is the case for $m = 0$, i.e. $L^p(\mathcal{D})$.

Figure 1: The concluding open questions in arXiv:2209.02703, PDF page 29.

2 Idea of the proof

Forget the derivatives and push the Sobolev Gaussian measure into $L^p(\mathcal{D})$. Proposition 2.9 realizes that pushforward as the law of the sample classes of a jointly measurable Gaussian process. Those

classes are supported on the Sobolev image. Although the image need not be closed in L^p , the Lusin–Souslin theorem makes it Borel and makes the inverse embedding Borel. Applying that inverse recovers exactly the original Sobolev-valued law.

3 An abstract realization theorem

Theorem 1. *Let $1 \leq p < \infty$, let $\mathcal{D} \subset \mathbb{R}^d$ be open, and let X be a separable Banach space equipped with a continuous injective linear map*

$$J : X \longrightarrow L^p(\mathcal{D}).$$

Assume the L^p -realization statement of Proposition 2.9 of [1]. Then every Gaussian Borel probability measure μ on X is induced by a jointly measurable Gaussian process $(U(x))_{x \in \mathcal{D}}$: its sample-class map is almost surely in $J(X)$, the map

$$\omega \longmapsto J^{-1}[U_\omega] \in X$$

is Borel measurable after an arbitrary definition on a null set, and its law is μ .

Proof. First suppose that μ is centered and put $\nu = J_*\mu$. This is a centered Gaussian measure on $L^p(\mathcal{D})$: for every $\ell \in (L^p)^*$, the pullback $\ell \circ J$ belongs to X^* , so the ℓ -marginal of ν is Gaussian.

By Proposition 2.9, there is a jointly measurable centered Gaussian process $U : \Omega \times \mathcal{D} \rightarrow \mathbb{R}$ such that its sample paths lie in L^p almost surely, the class map

$$F : \Omega \longrightarrow L^p(\mathcal{D}), \quad F(\omega) = [U_\omega],$$

is Borel, and $F_*\mathbb{P} = \nu$.

The spaces X and $L^p(\mathcal{D})$ are Polish. Since J is Borel and injective, the Lusin–Souslin theorem implies that

$$C := J(X)$$

is Borel in L^p , that $J^{-1} : C \rightarrow X$ is Borel, and, more generally, that $J(A)$ is Borel in L^p for every Borel $A \subset X$. Furthermore,

$$\mathbb{P}(F \in C) = \nu(C) = \mu(X) = 1.$$

Define $G = J^{-1} \circ F$ on this full-probability event and define it arbitrarily on its complement. Then $G : \Omega \rightarrow X$ is Borel. For every Borel $A \subset X$,

$$\mathbb{P}(G \in A) = \mathbb{P}(F \in J(A)) = \nu(J(A)) = \mu(A).$$

Thus G has law μ , and $F = J(G)$ almost surely. In particular, the sample class of U lies in $J(X)$ almost surely.

For a noncentered Gaussian measure, its mean a belongs to X because X is separable. Apply the centered case to the translate of μ by $-a$, choose a measurable representative $\widehat{J(a)}$ of $J(a)$, and replace $U(\omega, x)$ by $U(\omega, x) + \widehat{J(a)}(x)$. Joint measurability and Gaussianity are preserved, and the resulting X -valued law is μ . $\square$

Corollary 2. *Let $\mathcal{D} \subset \mathbb{R}^d$ be open, let $m \in \mathbb{N}_0$, and let $1 \leq p < \infty$. Every Gaussian Borel probability measure on $W^{m,p}(\mathcal{D})$ is induced by a jointly measurable Gaussian process with almost surely $W^{m,p}$ sample classes, and the Sobolev-valued sample-class map has exactly the prescribed law.*

Proof. The Sobolev space is a separable Banach space for finite p , and the canonical map

$$J : W^{m,p}(\mathcal{D}) \hookrightarrow L^p(\mathcal{D})$$

is continuous and injective. Apply Theorem 1. $\square$

The result includes $p = 1$, beyond the range $1 < p < \infty$ of the main Sobolev theorems in the source, because its L^p -realization Proposition 2.9 is valid for $1 \leq p < \infty$. The argument does not cover $p = \infty$, where both separability and the cited realization theorem fail in general.

4 What nuclear norm cannot say about small balls

Question (i) is broad, but its suggested scalar invariant has an exact negative answer.

Proposition 3. *Nuclear norm alone does not determine norm-small-ball probabilities, even for Gaussian measures on a Hilbert Sobolev space.*

Proof. Let $H = W^{m,2}(\mathcal{D})$ have dimension at least two and choose orthonormal $f_1, f_2 \in C_c^\infty(\mathcal{D})$ (start with two independent test functions and apply Gram–Schmidt in H). With independent standard normal variables, set

$$X_1 = \xi f_1, \quad X_2 = \frac{\xi_1 f_1 + \xi_2 f_2}{\sqrt{2}}.$$

Their covariance operators on H are

$$Q_1 = f_1 \otimes f_1, \quad Q_2 = \frac{1}{2}(f_1 \otimes f_1 + f_2 \otimes f_2),$$

so both have trace and nuclear norm one. Nevertheless, for $\varepsilon > 0$,

$$\mathbb{P}(\|X_1\|_H \leq \varepsilon) = 2\Phi(\varepsilon) - 1 \sim \sqrt{\frac{2}{\pi}} \varepsilon,$$

whereas

$$\mathbb{P}(\|X_2\|_H \leq \varepsilon) = \mathbb{P}(\xi_1^2 + \xi_2^2 \leq 2\varepsilon^2) = 1 - e^{-\varepsilon^2} \sim \varepsilon^2.$$

The small-ball orders themselves differ. $\square$

Even the complete eigenvalue list of the covariance on the *underlying* L^2 space is insufficient for a Sobolev norm. On $\mathcal{D} = (0, 2\pi)$, let

$$e_0 = (2\pi)^{-1/2}, \quad e_N(x) = \pi^{-1/2} \sin(Nx).$$

Both are L^2 -unit vectors, so the L^2 -covariance operators of ξe_0 and ξe_N have the identical spectrum $\{1, 0, 0, \dots\}$. But

$$\|e_0\|_{H^1} = 1, \quad \|e_N\|_{H^1} = \sqrt{1 + N^2},$$

and hence their H^1 small-ball probabilities are respectively

$$2\Phi(\varepsilon) - 1 \quad \text{and} \quad 2\Phi\left(\frac{\varepsilon}{\sqrt{1 + N^2}}\right) - 1.$$

Thus the Sobolev geometry of the covariance eigenfunctions matters. By contrast, the entire spectrum of the covariance *on the Hilbert Sobolev space itself* does determine the norm-small-ball law through the distribution of $\sum_j \lambda_j \xi_j^2$.

5 Verification and novelty notes

The realization proof is measure-theoretic and uses no computation. The accompanying script checks the exact small-ball formulas at a sample value of ε ; those formulas are proved analytically above.

A bounded official-arXiv search through 11 August 2026 used the exact question, source title and author, and combinations of the phrases *Gaussian measure Sobolev process realization*, *Lusin–Souslin*, and *small ball nuclear norm*. It found the source but no later answer to these exact questions. This is not an exhaustive literature review.

References

- [1] I. Henderson, *Sobolev regularity of Gaussian random fields*, arXiv:2209.02703.